Part B – Machine Learning and Software Integration

- **Due** *11:59pm Sunday 22 Nov 2026 (Week 11)*
- **Contributes** 50% to your Assignment 2 result.
- **Group of 3-4 students (same team as Assignment 2A)**.

The Part B of Assignment 2 (A2B) requires you to work with your team to implement ML algorithms to train ML models for traffic prediction and integrate the traffic predictor with Part A to develop a fully functioned TBRGS.

In Part B of Assignment 2, the team can use any machine learning technique or combinations of them. You should take advantage of existing libraries such as PyTorch, tensorflow, Keras, etc. We will provide you with a small dataset from VicRoads for the city of Boroondara that contains only the traffic flow data (the number of cars passing an intersection every 15 minutes). You should use this dataset for the following purposes: training ML models for predicting traffic conditions and testing/evaluating the performance of your models. At the very least, your TBRGS system will need to include the two basic deep learning techniques LSTM and GRU. Your TBRGS system will also need to implement at least one other technique (to be identified by you and approved by your tutor) to train another machine learning (ML) model and give a comprehensive comparison between different models. Your program will at least need to implement the following features:

- At the minimum, the TBRGS will have to be able to train ML models using the Boroondara dataset and give meaningful predictions based on these models
- A GUI will be available for the user input, parameter settings and visualisation (and a configuration file for the defaults).

The Traffic-based Route Guidance Problem

In this problem, there are four main tasks for the team:

1. Implement data processing program to extract data from the given dataset and store it in appropriate data structures to enable ML models to be trained/tested;
2. Implement ML algorithms to train ML models for traffic flow prediction using the provided dataset.
3. Implement a travel time estimation for each edge on the map of the Boroondara area.
4. Integrate Part A of Assignment 2 with the TBRGS to replace the nodes by the intersections, the cost of each edge by the travel time and perform the calculation to find the top-k paths to travel from O to D for any given pair (O,D) of origin and destination.

System requirements

For this assignment, we expect that the team will implement data processing methods and various ML algorithms (including LSTM and GRU) to train ML models for traffic prediction. The team will need to use the given dataset for training and also testing the ML models to evaluate their performance.

The team will also need to port the programs you developed previously for Assignment 2A to enable it to search for optimal paths on the Boroondara map. The edge cost will need to be replaced by the predicted travel time and subsequently, the optimal paths can be calculated and returned.

The basic version of TBRGS will be for the Boroondara area. A user can specify the origin and destination of their trip as the SCATS site number (e.g. origin O = 2000 [intersection WARRIGAL_RD/TOORAK_RD] and destination D = 3002 [intersection DENMARK_ST/BARKERS_RD]). The system then returns up to five (5) routes from O to D with the estimated travel time along each route. To simplify the calculation, you can make a number of assumptions: (i) The speed limit on every link will be the same and set at 60km/h; (ii) the travel time from a SCATS site A to a SCATS site B can be approximated by a simple expression based on the accumulated volume per hour at the SCATS site B and the distance between A and B (We will provide a simplified way to convert from traffic flow to travel time; See the document **Traffic Flow to Travel Time Conversion v1.0.PDF** on Canvas); and (iii) there is an average delay of 30 seconds to pass each controlled intersection. Note that, the objective is not to better Google Maps but to utilise the AI techniques you have

learned (e.g., machine learning for forecasting traffic volume, graph-based search for optimal paths) to solve a real-world problem.

Report file

You must also include a report which has to be either in Microsoft Word or in PDF whose name is your Team ID (for example, 123.PDF) containing your report. The aim of this report is for you to summarise your understanding of the problem and the algorithms you have used to solve the problem. We are most interested in the insights you have gained, especially those **using data obtained by running your software**.

Report Details: The report must be between 8 and 10 pages (excluding cover page and TOC).

- **Cover page:** including details of all group members (i.e., full names and student IDs), signed by all members of the group. A statement of contribution is required to provide specific information about each team member's contributions to the assignment.
- **Table of contents (TOC)**.
- **Instructions:** Basic instructions of how to use your program. You can also include a **note** containing anything else you want to tell the marker, such as how to use the GUI version of your program, and something particular about your implementation.
- **Introduction:** Introduce the *Traffic Prediction and Traffic-based Route Guidance Problem* and concise information about the algorithms the team have implemented.
- **Features/Bugs/Missing:** Include a list of the features you have implemented. Clearly state if a required feature has not been implemented. Failure to do this will result in penalties. Include a list of any known bugs. Also, anything else you want to tell the marker, such as how to use the GUI version of your program, and something particular about your implementation.
- **Testing:** Provide an overview of the test cases you have created to test your program (either manually or automatically). Report the results of testing your program. By testing, we refer to both the evaluation of the ML models and the software testing you have conducted to ensure that the software works well.
- **Insights:** Present and discuss the qualities of different ML algorithms used in your assignment. Which algorithms are better and why? **Use data collected to support your points.**
- **Research (if applicable):** If you managed to do some additional research to improve the program in some ways, please report it here.
- **Conclusion:** Conclude with a discussion about ML algorithms and their suitability to the problem and about the integration of different modules to produce the end-to-end TBRGS. Include thoughts about how you could improve performance.
- **Acknowledgements/Resources:** Include in your report a list of the resources you have used to create your work. A simple list of URL's is not enough. Include with each entry a basic description of how the person or website assisted you in your work.
- **References:** Cite the sources you referred to in your Assignment (implementation, report, etc.)

Tips:

- All figures and tables need to be properly captioned with sensible descriptions.
- Report presentation should include header/footer information (pages numbers, etc.)

Marking Scheme:

Requirements (or equivalent sections)	***Mark***
Implement appropriate data processing programs to extract data from the provided dataset and store data in appropriate data structures to enable ML models to be trained & tested	**9**
Implement at least 3 ML algorithms (including LSTM and GRU and an algorithm of your choice) for traffic flow prediction.	**7(x3)**
Conduct comprehensive evaluation to compare different ML models	**15**
Integration of Part A (**A2A**) & Part B to develop an end-to-end software for TBRGS	**15**

Testing: At least 15 test cases have been created to cover different problem scenarios. Test results have been checked and documented.	**10**
Report: Clear and provide sufficient information about the problem and your algorithm/solution AND the insights you have obtained through your program.	**15**
Research: If you show some initiatives in researching about the problem and solutions, or carrying out extensive tests to provide interesting data about the algorithms, or getting some clever optimization, etc. with a well-written **Research** section in the report to discuss and demonstrate these initiatives	**15**
Total	**100**
You need to follow good programming practice (e.g., well-designed, well-structure codes with clear and helpful comments). Failure to do so get penalty.	**Up to -10**

For each individual student: (the maximum mark a student can get for Assignment 2B is 100 and the minimum mark is 0)	***Mark***
Failure to provide in class updates of the assignment progress to your tutor.	**Up to -60**
Failure to contribute to the work performed by the team or to collaborate with other team members to achieve the desired outcomes for the project.	**Up to -60**

Notes:

1. A team may achieve a very high mark for A2B but a team member can still receive low mark because the student does not come to the class to provide assignment progress updates to their tutor or the student does not contribute to the assignment or both.
2. Since A2B contributes 50% to Assignment 2, the maximum mark you will see on Canvas for Assignment 2 Part B is 50, not 100.

Some ideas for research initiatives

- The standard version of the system only deals with a small area of Melbourne (the city of Boroondara) and uses a very small dataset (traffic volume data for the month of October 2006). Under this research initiative, you can look at more comprehensive datasets from VicRoads for the whole Victoria in multiple years. Data processing for large amount of data will be a challenge.

- The provided dataset may not provide you with details you need. You may want to consider other sources of open data. A good dataset can be found here: https://discover.data.vic.gov.au/dataset/traffic-signal-volume-data
  Some other popular data sources are: https://github.com/mas-dse-c6sander/DSE_Cohort2_Traffic_Capstone/wiki/PeMS-Data-Information and https://www.data.gov.uk/dataset/9562c512-4a0b-45ee-b6ad-afc0f99b841f/highways-england-network-journey-time-and-traffic-flow-data. Under this research initiative, you can choose one of these datasets or a good data source you have access to (and get your tutor's approval) and extend this project to deal with the selected network and dataset.

- Visualising your system predictions and route recommendations. Inspired by Google Maps Traffic, can you build something similar using open source resources such as OpenStreetMap?
  Note: *For some reason, the Latitude and Longitude of the SCATS sites do not map correctly to the actual intersections on Google Maps, you'll have to make adjustments to overcome this issue.*

- For other ideas, please feel free to do the research yourselves and discuss with your tutors to get their approval first before doing the research component.